

# CSE 6392 - 001: Advanced Topic: Seminar in LLM Security, Privacy & Ethics

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Office Hours: Thursday, 9 -11 AM, ERB 307

Class Format: Weekly seminar, 3 hrs, Discussion + Student Presentations

**Prerequisites:** Graduate-level background in ML, security and privacy (or instructor permission)

*As the instructor for this course, I reserve the right to adjust the class schedule or content in any way that serves the educational needs of the students enrolled in this course. — Faysal Hossain Shezan*

## Course Information

Time of Class Meetings: Wednesday, 4:00 PM - 6:50 PM

Place of Class Meetings: LS 101

## Course Description

This seminar examines the security, privacy, and ethical implications of large language models (LLMs) and related AI technologies. We begin with foundational adversarial machine learning concepts, then progress into contemporary vulnerabilities in LLMs (jailbreaking, prompt injection, agentic AI), modalities (deepfakes), platform implications (social media, software security), and finish with ethics and governance. Students engage deeply with cutting-edge research papers, lead seminar discussions, and produce a final project exploring a novel problem or defense mechanism.

## Technology Requirements

Assignments and slides are posted/need to upload on Canvas. You should submit your assignments on Canvas. All the classes will be in-person.

## Learning Outcomes

- Critically analyze adversarial attacks and defenses in LLM settings.
- Identify and categories novel vulnerabilities in LLM-based systems (e.g., jailbreaks, prompt-injection, agentic behaviors).
- Design or critique guardrail/mitigation strategies for LLMs and AI systems.
- Assess the social, platform, and governance implications of AI deployment on privacy, misinformation, and ethics.
- Conduct a small research project (experiment, or design proposal) and present findings in writing and orally.

## Weekly Schedule and Content

| Date       | Topic                                                                                        | Paper 1 title                                                                                                                                       | Paper 1 & Presenter        | Paper 2 title                                                                                                                                | Paper 2 & Presenter                  | Deliverables & Class Activity                           |
|------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------|
| 08/19/2026 | <b>Week 1: Introduction and Course Overview - Landscape of ML Security, LLMs, and Ethics</b> | Course Overview                                                                                                                                     |                            | Malla: Demystifying Real-world Large Language Model Integrated Malicious Services<br><i>USENIX Security 2024</i>                             | <a href="#">Open paper</a><br>Faysal |                                                         |
| 08/26/2026 | <b>Week 2: LLM Model Context Protocol Security</b>                                           | IsolateGPT: An Execution Isolation Architecture for LLM-Based Agentic Systems<br><i>NDSS 2025</i>                                                   | <a href="#">Open paper</a> | Demystifying RCE Vulnerabilities in LLM-Integrated Apps<br><i>ACM CCS 2024</i>                                                               | <a href="#">Open paper</a>           | Class: Idea Presentation                                |
| 09/02/2026 | <b>Week 3: Vision-Language Model Security and Privacy</b>                                    | Membership Inference Attacks Against Vision-Language Models<br><i>USENIX Security 2025</i>                                                          | <a href="#">Open paper</a> | Mirage in the Eyes: Hallucination Attack on Multi-modal Large Language Models with Only Attention Sink<br><i>USENIX Security 2025</i>        | <a href="#">Open paper</a>           |                                                         |
| 09/09/2026 | <b>Week 4: LLM Security and Privacy Foundations</b>                                          | Did the Neurons Read your Book? Document-level Membership Inference for Large Language Models<br><i>USENIX Security 2024</i>                        | <a href="#">Open paper</a> | What Was Your Prompt? A Remote Keylogging Attack on AI Assistants<br><i>USENIX Security 2024</i>                                             | <a href="#">Open paper</a>           |                                                         |
| 09/16/2026 | <b>Week 5: Jailbreaking of LLMs</b>                                                          | Don't Listen To Me: Understanding and Exploring Jailbreak Prompts of Large Language Models<br><i>USENIX Security 2024</i>                           | <a href="#">Open paper</a> | Making Them Ask and Answer: Jailbreaking Large Language Models in Few Queries via Disguise and Reconstruction<br><i>USENIX Security 2024</i> | <a href="#">Open paper</a>           |                                                         |
| 09/23/2026 | <b>Week 6: Prompt Injection and Retrieval-Augmented Systems</b>                              | PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models<br><i>USENIX Security 2025</i>                 | <a href="#">Open paper</a> | Optimization-based Prompt Injection Attack to LLM-as-a-Judge<br><i>ACM CCS 2024</i>                                                          | <a href="#">Open paper</a>           | Canvas: Project Proposal<br>Class: Project Presentation |
| 09/30/2026 | <b>Week 7: Agentic AI and Multi-Step Reasoning Attacks</b>                                   | Make Agent Defeat Agent: Automatic Detection of Taint-Style Vulnerabilities in LLM-based Agents<br><i>USENIX Security 2025</i>                      | <a href="#">Open paper</a> | Cloak, Honey, Trap: Proactive Defenses Against LLM Agents<br><i>USENIX Security 2025</i>                                                     | <a href="#">Open paper</a>           | Class: Weekly Project Update                            |
| 10/07/2026 | <b>Week 8: Guardrails, Alignment, and Monitoring</b>                                         | JBSHield: Defending Large Language Models from Jailbreak Attacks through Activated Concept Analysis and Manipulation<br><i>USENIX Security 2025</i> | <a href="#">Open paper</a> | Refusal Is Not an Option: Unlearning Safety Alignment of Large Language Models<br><i>USENIX Security 2025</i>                                | <a href="#">Open paper</a>           | Class: Weekly Project Update                            |

| Date       | Topic                                                                  | Paper 1 title                                                                                                                          | Paper 1 & Presenter        | Paper 2 title                                                                                                                                                          | Paper 2 & Presenter        | Deliverables & Class Activity                     |
|------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------|
| 10/14/2026 | <b>Week 9: Deepfakes and Multimodal Threats</b>                        | Chimera: Creating Digitally Signed Fake Photos by Fooling Image Recapture and Deepfake Detectors<br><i>USENIX Security 2025</i>        | <a href="#">Open paper</a> | AudioMarkNet: Audio Watermarking for Deepfake Speech Detection<br><i>USENIX Security 2025</i>                                                                          | <a href="#">Open paper</a> | Class: Weekly Project Update                      |
| 10/21/2026 | <b>Week 10: Social Media, Platform Abuse, and Misinformation</b>       | HateBench: Benchmarking Hate Speech Detectors on LLM-Generated Content and Hate Campaigns<br><i>USENIX Security 2025</i>               | <a href="#">Open paper</a> | SoK: Machine Learning for Misinformation Detection<br><i>USENIX Security 2025</i>                                                                                      | <a href="#">Open paper</a> | Class: Weekly Project Update                      |
| 10/28/2026 | <b>Week 11: Software Security, LLMs in DevOps, and Code Generation</b> | Do Users Write More Insecure Code with AI Assistants?<br><i>ACM CCS 2023</i>                                                           | <a href="#">Open paper</a> | An LLM-Assisted Easy-to-Trigger Backdoor Attack on Code Completion Models: Injecting Disguised Vulnerabilities against Strong Detection<br><i>USENIX Security 2024</i> | <a href="#">Open paper</a> | Class: Weekly Project Update                      |
| 11/04/2026 | <b>Week 12: Ethics, Governance, and Responsible AI Deployment</b>      | Generated Data with Fake Privacy: Hidden Dangers of Fine-tuning Large Language Models on Generated Data<br><i>USENIX Security 2025</i> | <a href="#">Open paper</a> | Exploring User Security and Privacy Attitudes and Concerns Toward the Use of General-Purpose LLM Chatbots for Mental Health<br><i>USENIX Security 2025</i>             | <a href="#">Open paper</a> | Class: Weekly Project Update                      |
| 11/11/2026 | <b>Week 13: Wrap-Up, Reflection, and Future Directions</b>             | Chasing Shadows: Pitfalls in LLM Security Research<br><i>NDSS 2026</i>                                                                 | <a href="#">Open paper</a> | From Large to Mammoth: A Comparative Evaluation of Large Language Models in Vulnerability Detection<br><i>NDSS 2025</i>                                                | <a href="#">Open paper</a> | Class: Weekly Project Update                      |
| 11/18/2026 | <b>Presentation</b>                                                    |                                                                                                                                        |                            |                                                                                                                                                                        |                            | Canvas: Final Report<br>Class: Final Presentation |
| 11/25/2026 | <b>Thanksgiving Break</b>                                              |                                                                                                                                        |                            |                                                                                                                                                                        |                            |                                                   |

## Assessments and Grading

| Assessment Component                 | Weight |
|--------------------------------------|--------|
| Participation and Seminar Discussion | 10%    |
| Short Assignments                    | 10%    |

| Assessment Component                               | Weight    |
|----------------------------------------------------|-----------|
| Seminar Leadership                                 | 20%       |
| Final Project: Paper, Presentation, and Milestones | 50%       |
| Attendance                                         | 10%       |
| Bonus Opportunities                                | Up to 10% |

## Final Project

Students will complete a semester-long research project in teams of three. The class is expected to have approximately eight teams. Students are encouraged to propose original project ideas related to the course; the instructor will also provide several well-scoped project ideas.

The project should address a meaningful research problem and make a clear contribution to LLM security. Projects may involve designing a new system or technique, evaluating an existing approach, reproducing and extending prior research, developing a benchmark or dataset, or conducting a rigorous empirical study.

The final project is worth 50% of the course grade and includes the proposal, milestone updates, progress report, final paper, and presentation. Projects with the potential to become conference or workshop submissions are especially encouraged.

## Deliverables

1. Submit a project proposal describing the research problem, motivation, research questions, proposed methodology, expected security contributions, evaluation metrics, and project timeline.
2. Provide brief weekly progress updates and participate in periodic milestone meetings.
3. Submit a written progress report containing preliminary results and a draft of the related-work section.
4. Submit a research paper of no more than six pages in a standard two-column conference format, excluding references.
5. Deliver a 15-minute final presentation followed by approximately five minutes of questions and discussion.

Team members are expected to contribute equitably to the project. Individual grades may be adjusted based on peer evaluations, demonstrated contributions, and participation in project meetings and presentations.

## Project Timeline

### Idea Presentations

**Week 2:** Students will briefly present their proposed ideas in class. Approximately 10 minutes will be allocated to each presentation and discussion. By the end of the class, select the project idea they plan to pursue.

- The proposed research question or problem
- The motivation for addressing the problem
- The general technical or empirical approach
- The anticipated contribution to computer security

### Project Refinement Meetings

**Week 3:** [If needed] the instructor will meet with each team to discuss the proposed project, evaluate its scope and feasibility, and refine its research questions, methodology, and evaluation plan.

### Project Proposal

**Week 6:** Each team must submit a three-page final project proposal containing:

- The problem statement and motivation
- Research questions
- Hypotheses, when applicable
- Proposed methodology
- Datasets, tools, systems, or experimental environments
- Evaluation metrics and baselines
- Expected contributions
- Potential risks or limitations
- A timeline containing specific milestones, checkpoints, and deliverables
- A description of what the team expects to accomplish by the end of the semester

### Project Presentations

**Week 6:** Students will briefly present their project proposal in class. Approximately 10 minutes will be allocated to each presentation and discussion. By the end of the class, select the project idea they plan to pursue.

### Weekly Progress Reports

**Week 7-12: Every week in class:** Each team will provide an approximately five-minute oral progress report. The report should summarize:

- Work completed during the previous week
- Preliminary findings or technical accomplishments
- Problems, risks, or blockers
- Planned activities for the following week
- Progress relative to the approved project timeline

These reports are intended to identify problems early and keep projects on schedule.

### Final Report

**Week 13:** Each team must submit a research paper of no more than six pages in a standard two-column conference format, excluding references. The paper should include:

- Abstract
- Introduction and motivation
- Research questions or hypotheses
- Related work
- Methodology or system design
- Experimental setup
- Results
- Discussion
- Limitations and threats to validity

- Conclusions and future work
- References

You can follow this latex template from [Association for Computing Machinery \(ACM\) - Large 2-Column Format Template - Overleaf, Online LaTeX Editor](#)

### Final Presentation

**Week 14:** Each team will deliver a 15-minute final presentation followed by approximately 5 minutes of questions and discussion. The presentation should clearly explain:

- The research problem and motivation
- Research questions
- Proposed approach or system
- Experimental design
- Results
- Limitations
- Security implications
- Conclusions and future work

Every team member must participate meaningfully in the presentation and question-and-answer session.

Grading For Project:

- Project proposal presentation      5%
- Weekly project progress updates.    30%
- Final presentation                      30%
- Final project report                      35%

### Seminar Leadership

Most class meetings will include two students as seminar leaders. Each seminar-lead presentation will occupy approximately 30 minutes, followed by approximately 25 minutes of instructor-moderated and student-led discussion. A short break may be provided between presentation sessions.

Each student will serve as a discussion leader for one assigned research paper.

For each assigned paper, the seminar leader must prepare a professional 25-minute presentation that covers:

- The problem and motivation
- The paper's research questions
- The main technical ideas and contributions
- The system, methodology, or experimental design
- Datasets, baselines, and evaluation metrics
- Major results
- Strengths and weaknesses
- Limitations and threats to validity

- Connections to other papers and course topics
- Potential extensions and open research questions
- **Have you used any LLM/AI for solving any part of this task? If yes, how have you ensured that you followed the right way (e.g., ethical, responsible, accountable)? If not, explain why. Meta data of the LLM(s) that you used (such as, name, type, version).**

The seminar leader must also prepare at least 5 substantive discussion questions. These questions should encourage critical analysis rather than simple factual recall. For example, questions may examine methodological choices, assumptions, experimental validity, alternative approaches, security implications, ethical concerns, or opportunities for future research.

Before the beginning of class, seminar leaders must upload their presentation slides as a PDF to the designated Student Slides folder in Canvas. Seminar leaders are responsible for using their prepared questions to guide the discussion. All other students are expected to participate actively and contribute thoughtful comments, questions, and critiques.

## Short Assignments: Paper Reviews

Students will read and discuss several research papers throughout the semester. All assigned readings must be completed before class so that students can participate meaningfully in seminar discussions.

Unless otherwise stated, students must submit a review of each assigned paper through Canvas by Sunday at 11:59 p.m. Reviews must be written in the student's own words and should demonstrate both comprehension and critical analysis.

### Section 1: Summary and Technical Analysis

- Have you used any LLM/AI for solving any part of this task?
  - If yes, how have you ensured that you followed the right way (e.g., ethical, responsible, accountable)? Meta data of the LLM(s) that you used (such as, name, type, version).
  - If not, explain why.

### Section 2: Summary and Technical Analysis

Briefly explain:

- The problem addressed by the paper and why it is important
- The paper's research questions or objectives
- The main ideas and technical contributions
- How the work differs from or improves upon prior research
- The methodology used to design or evaluate the proposed approach
- The datasets, baselines, metrics, and experimental settings
- The principal findings and conclusions

### Section 3: Strengths and Weaknesses

Identify and explain:

- Three important strengths of the paper
- Three important weaknesses or limitations of the paper

The analysis should focus primarily on the novelty and soundness of the approach, quality of the evaluation, realism of the assumptions and threat model, appropriateness of the baselines, reproducibility, limitations, and validity of the conclusions. Comments about writing or presentation should be included only when they materially affect the paper's clarity or technical credibility.

## Section 4: Discussion Questions and Future Directions

Provide:

- At least two substantive questions for class discussion
- One possible extension, alternative approach, or future research direction

Reviews that only summarize the paper without providing meaningful critical analysis will not receive full credit.

## Participation and Seminar Discussion

This is a discussion-based graduate seminar, and active preparation and participation are essential components of the course.

Students are expected to:

- Complete all assigned readings before class
- Arrive prepared to discuss the papers
- Respond thoughtfully when called upon
- Ask relevant questions
- Critically evaluate research methods and conclusions
- Contribute constructively to discussions led by classmates
- Respectfully consider alternative interpretations and perspectives

Cold calling may be used to invite students to answer questions or comment on assigned readings. Questions may assess comprehension, but they will also encourage students to evaluate assumptions, trade-offs, methodological decisions, and security implications. Research questions often do not have a single correct answer; therefore, students will be evaluated on the quality of their reasoning, use of evidence, and ability to engage constructively with the discussion.

Participation grades will reflect both the frequency and quality of contributions. Speaking frequently without demonstrating preparation or engaging with the substance of the discussion will not necessarily result in a high participation grade.

## Submission

- All the presentations and reports are due by Sunday 11:59 PM for that week.

## Attendance

Because presentations, discussions, and project updates are central to the course, regular attendance is required. Students should arrive on time and remain for the full class session.

Absences may be excused in accordance with university policy. Students are responsible for notifying the instructor when appropriate and completing any missed work. Repeated unexcused absences, late arrivals, or early departures may reduce the attendance grade and may also affect participation and seminar-leadership grades.

## Institution Information

UTA students are encouraged to review the below institutional policies and informational sections and reach out to the specific office with any questions. To view this institutional information, please visit the [Institutional Information](https://resources.uta.edu/provost/course-related-info/institutional-policies.php) page (https://resources.uta.edu/provost/course-related-info/institutional-policies.php), which includes the following policies, among others:

- Drop Policy
- Disability Accommodations
- Title IX Policy
- Academic Integrity
- Student Feedback Survey
- Final Exam Schedule

## Recording of Classroom and Online Lectures

Faculty maintain the academic right to determine whether students are permitted to record classroom and online lectures. Recordings of classroom lectures, if permitted by the instructor or pursuant to an ADA accommodation, may only be used for academic purposes related to the specific course. They may not be used for commercial purposes or shared with non-course participants except in connection with a legal proceeding.

## Plagiarism

All students are expected to take responsibility for their own work. Full or partial copying of another's work will be considered *plagiarism*. The instructor reserves the right to make decisions in such cases. Students found engaging in these activities may face penalties ranging from receiving a score of zero to a negative score for the assignment. Both the original author of the content and the individual who copied it will be held equally responsible and face the same consequences. Based on the situation, the instructor reserves the right to decide the penalty.

## Generative AI Use in This Course

The use of Generative AI (GenAI) in course assignments and assessments must align with the guidelines established by the instructor. Unauthorized use of GenAI could result in breaches of academic integrity. Instructors bear the responsibility of clearly delineating the permissible uses of GenAI in their courses, underscoring the importance of responsible and ethical application of these tools. The UTA Office of Community Standards articulates the university's stance on academic integrity and scholastic dishonesty. These standards extend to the use of GenAI. Unauthorized or unapproved use of GenAI in academic work falls within the scope of these policies and will be subject to the same disciplinary procedures.

## Additional Information

### Mandatory Face Covering Policy

*Face coverings are not mandatory; all students and instructional staff are welcome to wear face coverings while they are on campus or in the classroom.*

### Emergency Exit Procedures:

Should we experience an emergency event that requires evacuation of the building, students should exit the room and move toward the nearest exit. When exiting the building during an emergency, do not take an elevator but use the stairwells instead. Faculty members and instructional staff will assist students in selecting the safest route for evacuation and will make arrangements to assist individuals with disabilities.

**Academic Success Center:**

The Academic Success Center (ASC) includes a variety of resources and services to help you maximize your learning and succeed as a student at the University of Texas at Arlington. ASC services include supplemental instruction, peer-led team learning, tutoring, mentoring and TRIO SSS. Academic Success Center services are provided at no additional cost to UTA students. For additional information visit: [Academic Success Center](https://www.uta.edu/student-success/course-assistance) (https://www.uta.edu/student-success/course-assistance). To request disability accommodations for tutoring, please complete this [tutoring request form](https://www.uta.edu/student-success/course-assistance/tutoring/request) (https://www.uta.edu/student-success/course-assistance/tutoring/request).

**The IDEAS Center** (https://www.uta.edu/ideas/) (2nd Floor of Central Library) offers **FREE** [tutoring](#) and [mentoring](#) to all students with a focus on transfer students, sophomores, veterans and others undergoing a transition to UT Arlington. Students can drop in or check the schedule of available peer tutors at [www.uta.edu/IDEAS](http://www.uta.edu/IDEAS), or call (817) 272-6593.

**The English Writing Center (411LIBR):**

The Writing Center offers **FREE** tutoring in 15-, 30-, 45-, and 60-minute face-to-face and online sessions to all UTA students on any phase of their UTA coursework. Register and make appointments online at the [Writing Center](https://uta.mywconline.com) (https://uta.mywconline.com). Classroom visits, workshops, and specialized services for graduate students and faculty are also available. Please see [Writing Center: OWL](http://www.uta.edu/owl) (http://www.uta.edu/owl) for detailed information on all our programs and services.

**Librarian to Contact:**

Each academic unit has access to [Librarians by Academic Subject](http://www.uta.edu/library/help/subject-librarians.php) (http://www.uta.edu/library/help/subject-librarians.php) that can assist students with research projects, tutorials on plagiarism and citation references as well as support with databases and course reserves.

**Emergency Phone Numbers**

In case of an on-campus emergency, call the UT Arlington Police Department at **817-272-3003** (non-campus phone), **2-3003** (campus phone). You may also dial 911. Non-emergency number 817-272-3381.